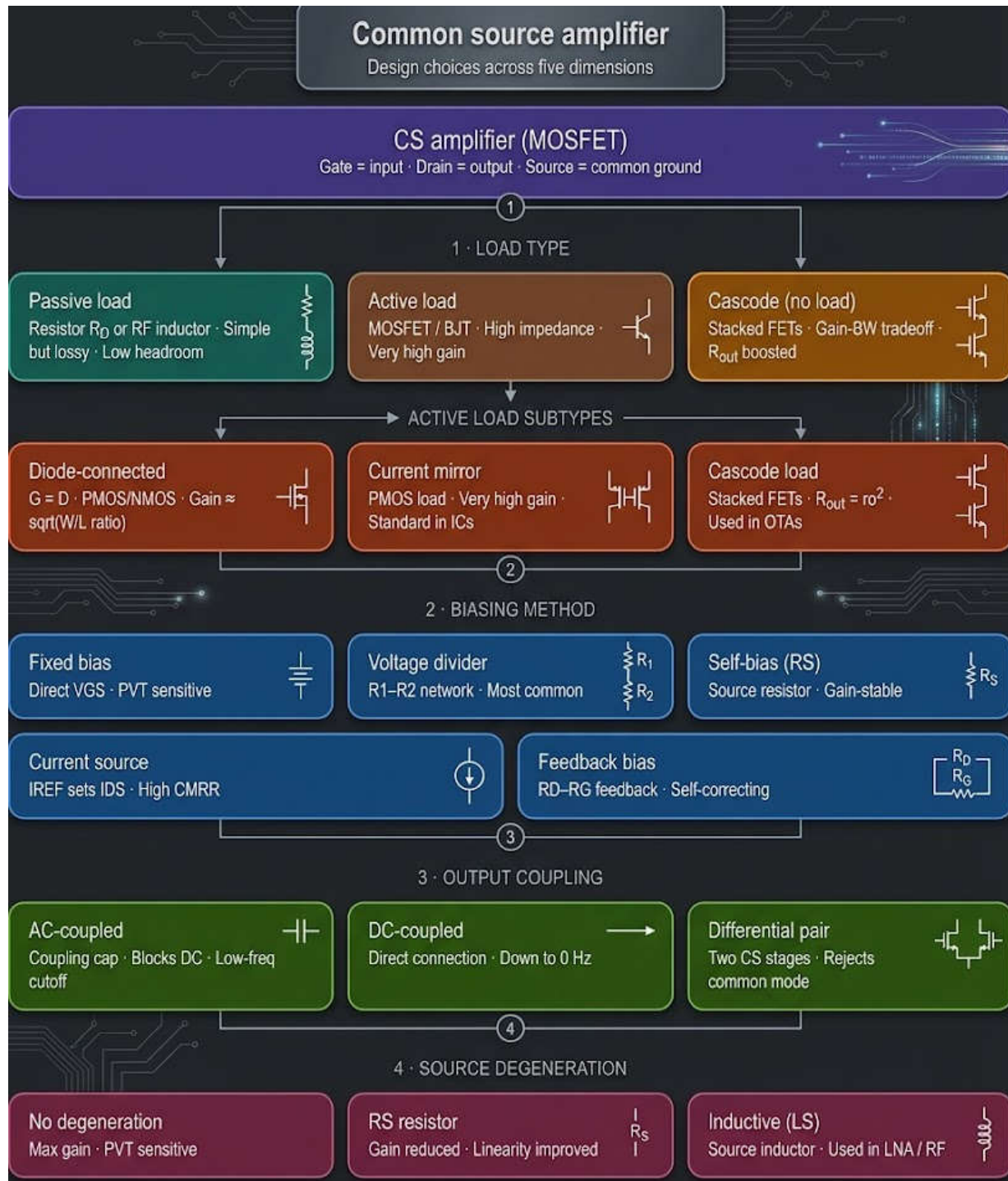


Common Source Amplifier Configurations

There are **many possible design configurations** for a common source (CS) amplifier. The key design choices revolve around what you use as the **drain load** and **biasing network**. Here's a full breakdown:



What is Active Load?

In MOSFET amplifier circuits, instead of a passive resistor, the active component like a MOSFET or MOSFET circuit (Current Mirror) is used to increase the gain of the amplifier. This active component or the active circuit is known as the Active Load.

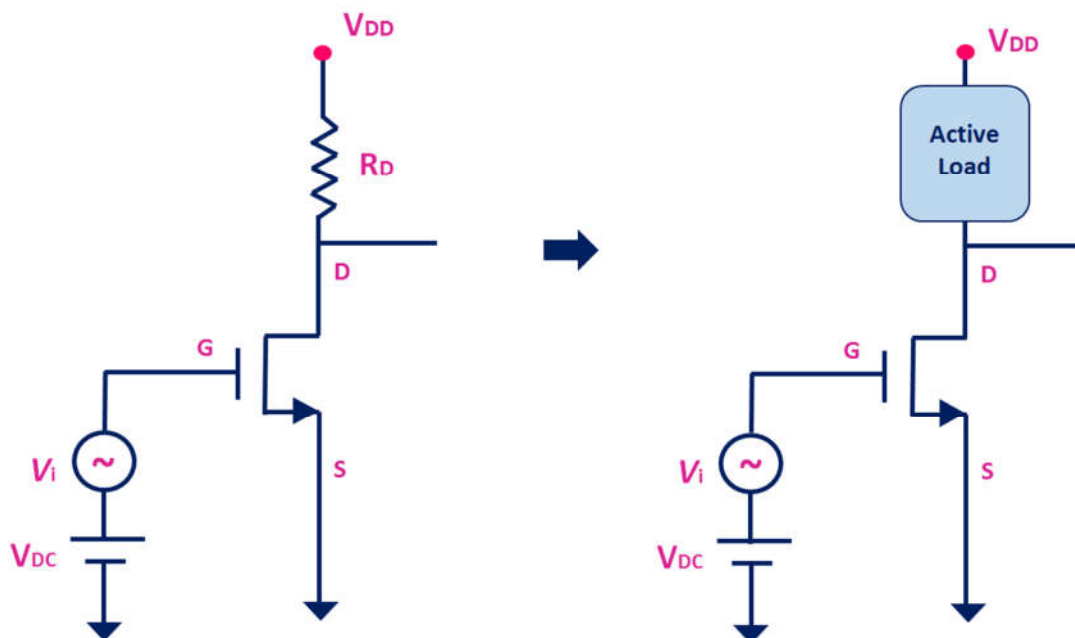


Fig. 1 MOSFET Amplifier with Active Load

Why is Active-Load used in the circuit design?

The Active Load is usually used in the integrated circuits where the size and power consumption are major constrain. Also, in the integrated circuits, fabricating a resistor requires a lot of space. And therefore, instead of a resistor, the active load is used. The active load also helps in increasing the voltage gain of the amplifier. To understand this point, let's take the example of a simple common source amplifier with the passive load.

The gain of the common source amplifier is

$$|A_v| = g_m * (R_D \parallel r_o) \text{ where } r_o \text{ is output resistance}$$

$$\text{If } r_o \gg R_D \text{ then } |A_v| \approx g_m * R_D$$

To increase the gain, either R_D or g_m needs to be increased. But as R_D increases, the voltage drop across R_D also increases and hence, the available voltage at the drain terminal reduces. At one stage, the MOSFET may come out of the saturation.

Similarly, by increasing the drain current I_D , the transconductance (g_m) can be increased. But as the drain current increases, the power dissipation in the circuit increases. Also, with the increase in the drain current, the voltage drops across the drain resistor will increase. And at one point, MOSFET may come out of the saturation.

Of course, by increasing the power supply voltage, the MOSFET can be kept in the saturation region, and it is possible to increase the gain by some extent. But that is not an option in the modern

integrated circuits where the supply voltage range is shrinking day by day. All these problems can be eliminated using the active load.

Feature	Passive (resistor)	Active (MOSFET)
<i>Element type</i>	<i>Non-amplifying, dissipative</i>	<i>Has its own g_m, can amplify</i>
<i>Small-signal impedance</i>	<i>Fixed = R_D</i>	<i>Very high: $r_o = 1/\lambda \cdot I_D$</i>
<i>DC voltage drop</i>	<i>Large ($I_D \cdot R_D$)</i>	<i>Small ($V_{DSsat} \approx 100\text{--}200\text{mV}$)</i>
<i>Gain achievable</i>	<i>Moderate ($\sim 10\text{--}30$)</i>	<i>Very high ($\sim 100\text{--}1000$)</i>
<i>Used in</i>	<i>Discrete, RF, simple stages</i>	<i>All VLSI/IC op-amps, OTAs</i>
<i>Power efficiency</i>	<i>Low</i>	<i>High</i>